Electricity and Electrical Safety

Programmable Logic Controllers

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Agenda

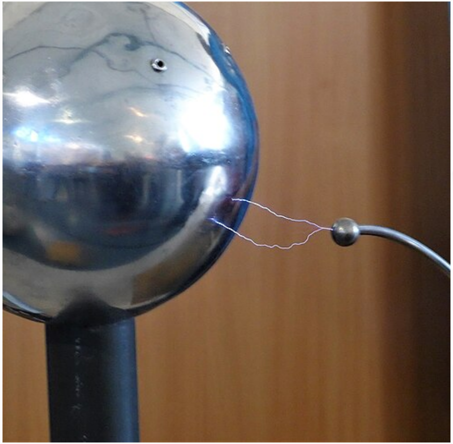
Electricity Review

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Electricity Review

Electricity



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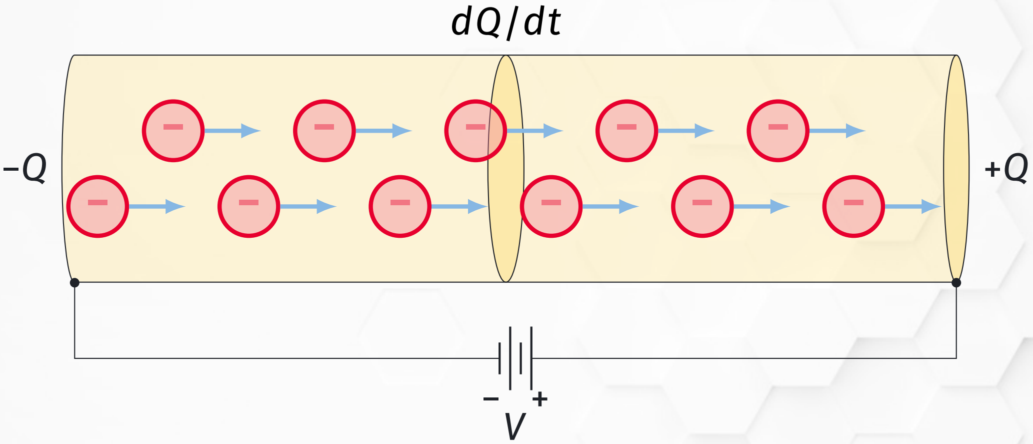
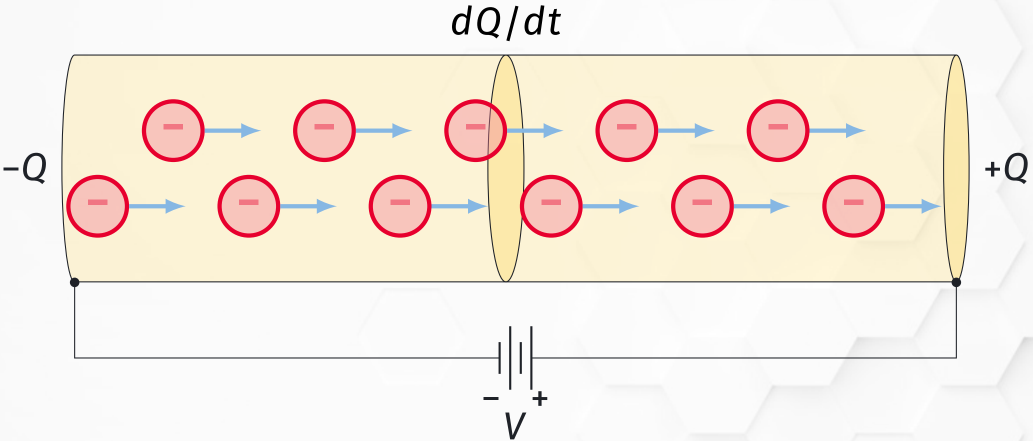
Electricity is a form of energy resulting from the existence of charged particles (protons or electrons)

Current Defined



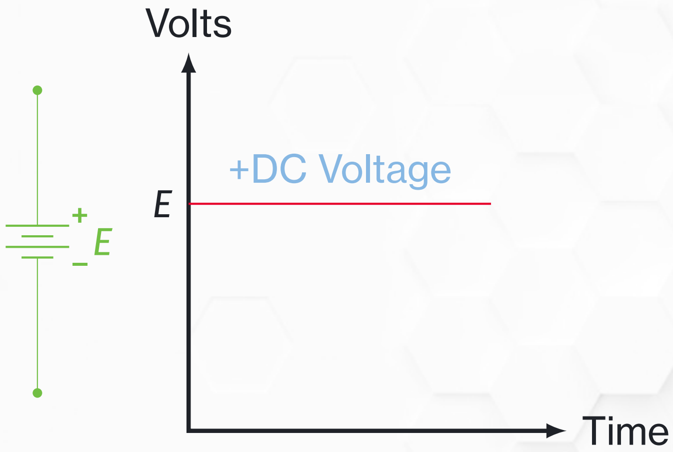
Current is the *flow rate* (speed) of electric charge through a conductor

Ampere Defined



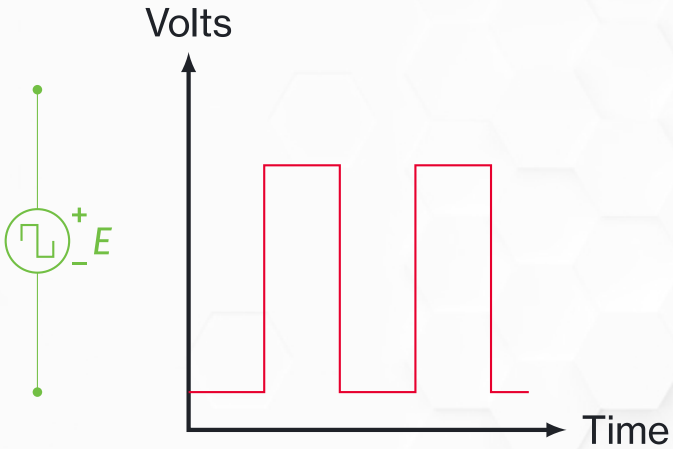
One *ampere* is 1 coloumb of charge (6.24×10^{18} electrons) flowing past a single point in 1 second

Direct Current Defined



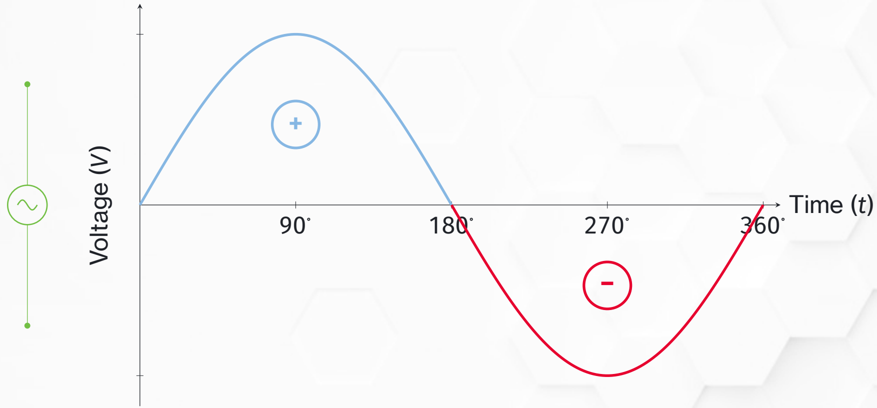
Direct current is an electric current that maintains its polarity (flows in only one direction)

Pulsed DC



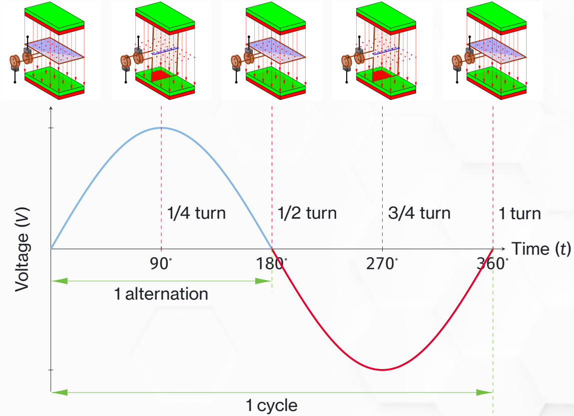
Direct current **does not** need to be a steady voltage – it can be *pulsed*. As long as it maintains polarity it is still DC.

Alternating Current Defined



Alternating current is an electric current that reverses polarity. It **does not** need to be a sine wave, but the sine wave is the most common.

AC Voltage - Sine Wave



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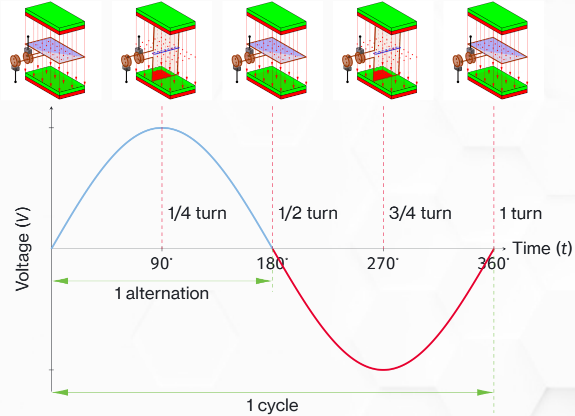
The sine wave is caused by the physical layout of an AC generator – as the coil spins inside the magnetic field it cuts the field in different directions, leading to a reverse in polarity

AC Voltage - Alternation



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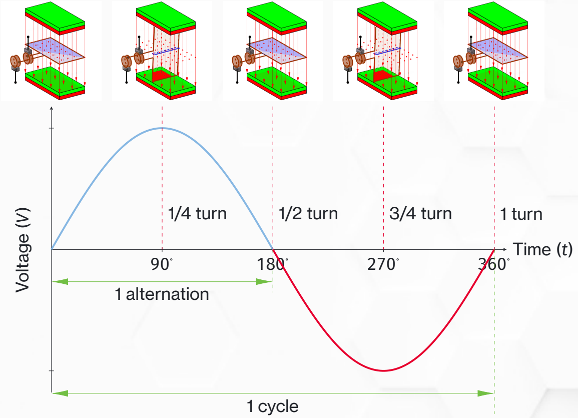
An *alternation* is 1/2 of a rotation of the coil–the point at which the polarity changes

AC Voltage - Cycle



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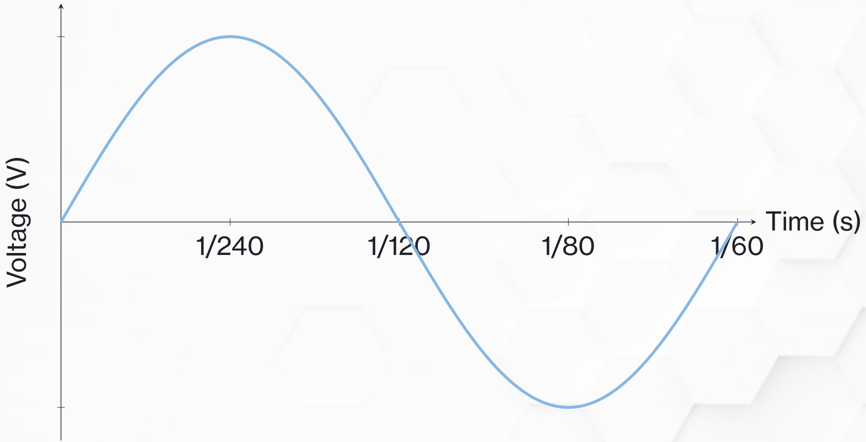
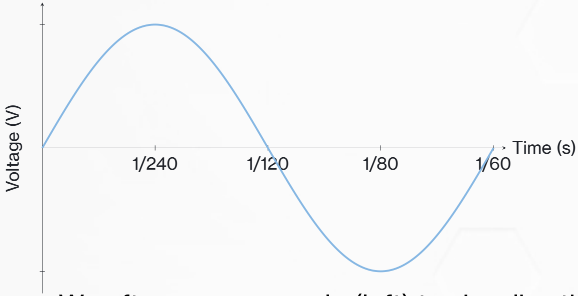
A *cycle* is a full rotation of the coil

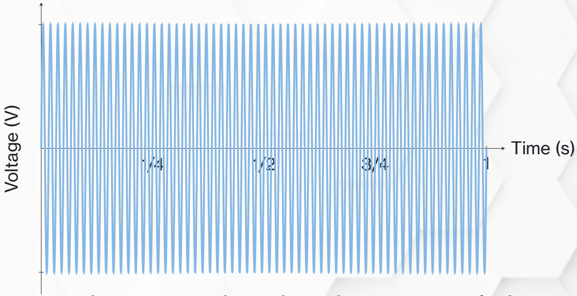
AC Voltage - Frequency



The frequency is the number of cycles per second (Hz)–the U.S. standard is 60 Hz (60 cycles per second)

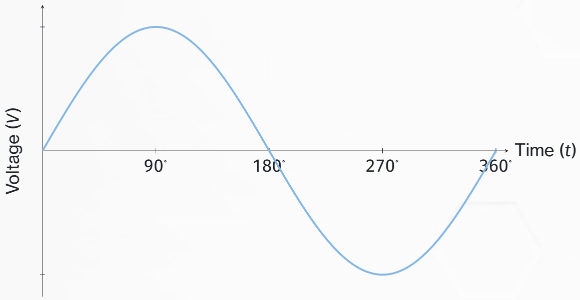
Visualizing Frequency

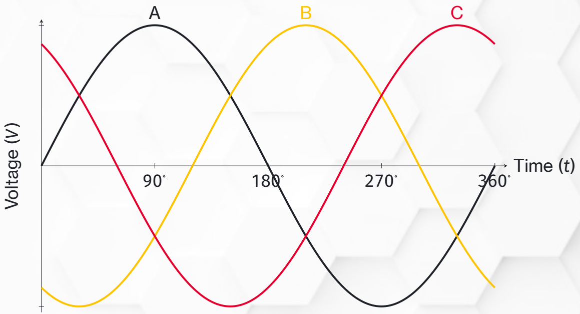




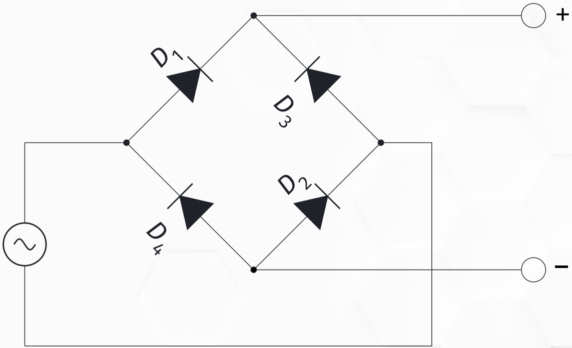
We often use one cycle (left) to visualize the curve, but remember that that one cycle is happening 60 times every second (right)

AC Voltage - Phases



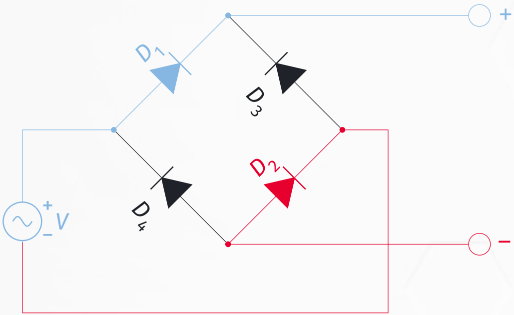


Bridge Rectifier

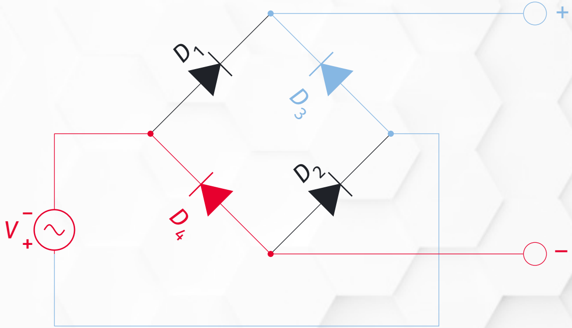


A *bridge rectifier* is commonly used in controls applications to convert AC to DC

Bridge Rectifier



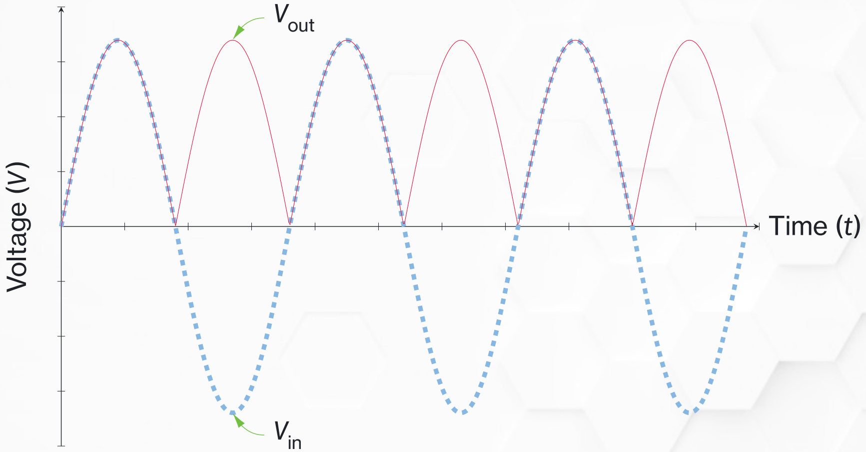
Positive AC Cycle



Negative AC Cycle

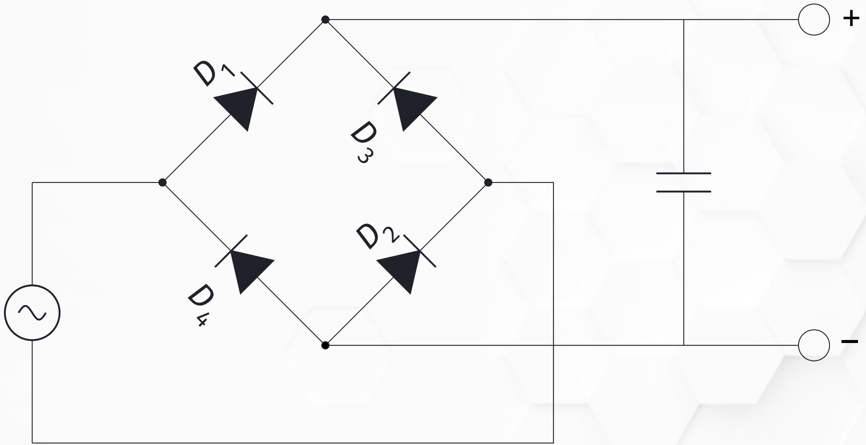
As the polarity of the AC voltages changes the the rectifier forces the output to the same polarity

Rectified Alternating Current



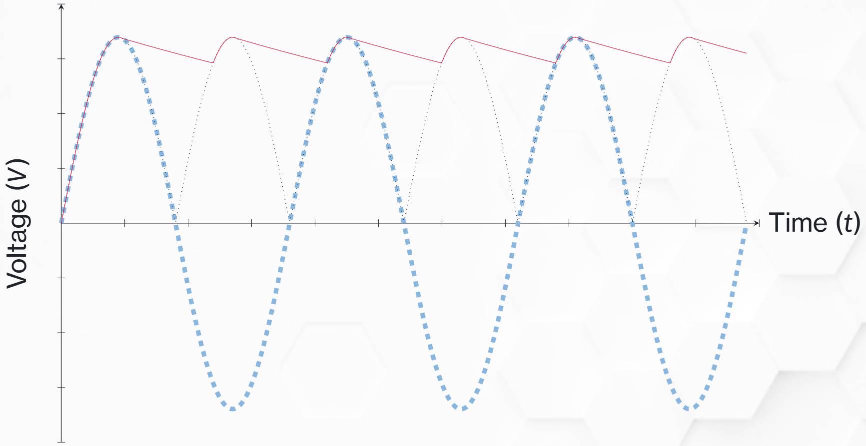
The rectifier redirects current so the output is always the same polarity, but it *does not* affect the voltage value–the resulting wave is DC, but not very useful DC

Rectified Alternating Current



Smoothing capacitors are added to help maintain the voltage level between cycles, which helps keep the DC at a more consistent value

Rectified Alternating Current



The result is a fairly smooth DC wave–the *ripple* will always be there, but can be tweaked by changing the capacitor value